Customer Personality Analysis
Team : ML8

Team Members:

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Business Objectives

Client: ABC Supermart

Analyze customer purchasing behavior

Identify distinct customer segments

Personalized / targeted marketing campaigns

Improved KPIs
Campaign effectiveness, Customer retention, and long term revenue growth

Images source : https://www.flaticon.com/

Analyze Customer Purchasing Behaviour

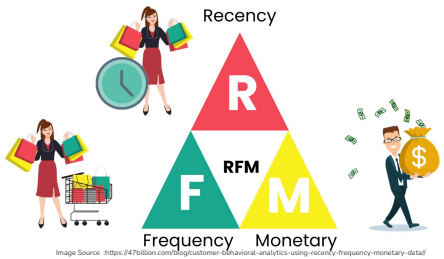
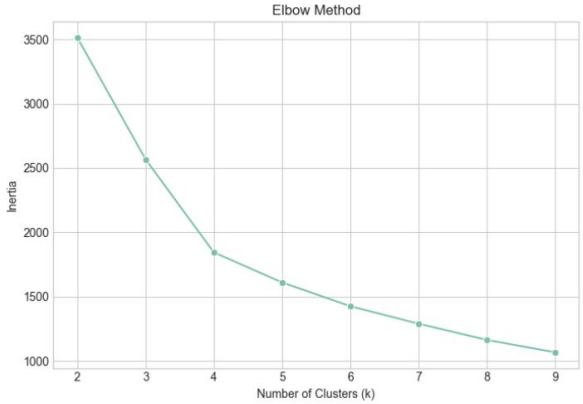


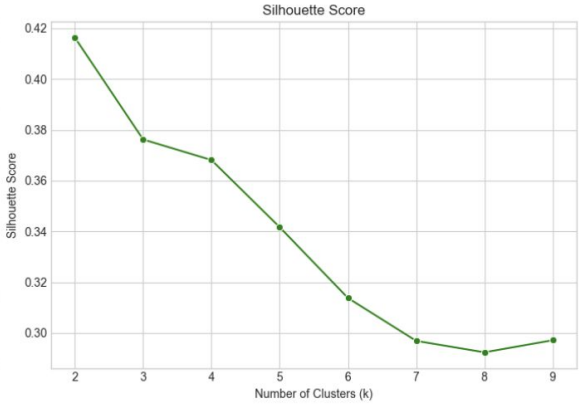
Image Source :https://47billion.com/blog/customer-behavioral-analytics-using-recency-frequency-monetary-data/

Features used for model training

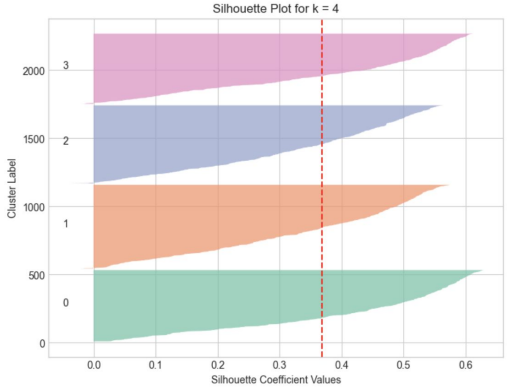
K Means Clustering

- Unsupervised Task
- RFM features are numerical
- Widely used for customer segmentation

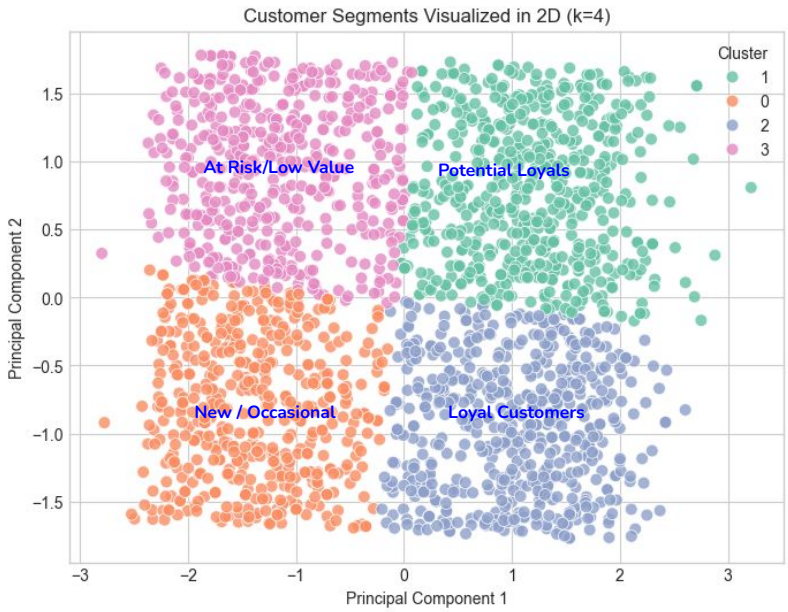




Determined the optimal number of clusters by comparing multiple K values and maximizing the Silhouette Score, ensuring well-separated and internally cohesive clusters



Identify Distinct Customer Segments



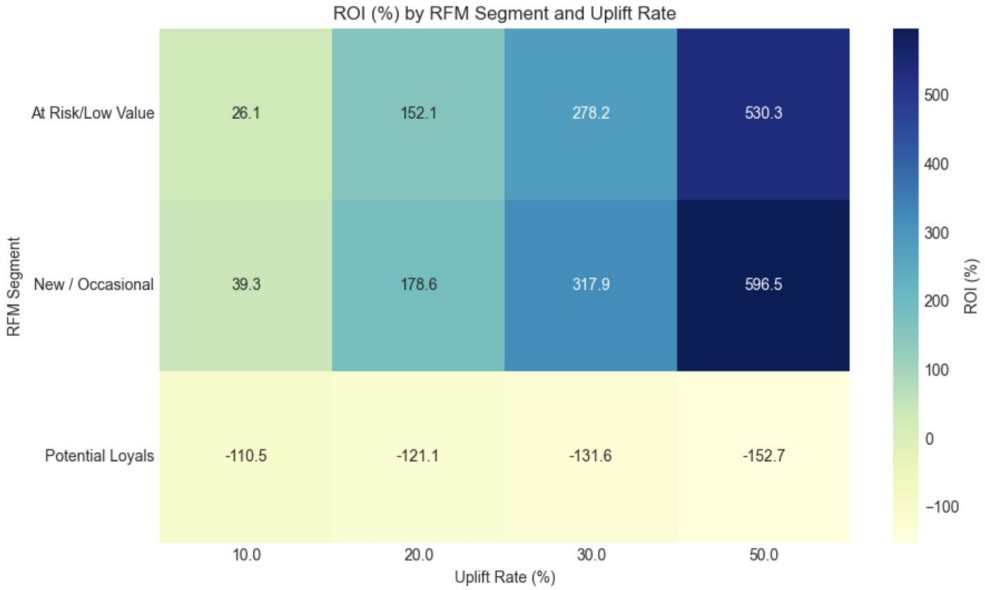
Cluster	Segment Name	No. of Customers	General Characteristics
Cluster 0	New / Occasional	524	Low Recency, Low Frequency, Low Monetary
Cluster 1	Potential Loyals	616	High Recency, High Frequency, High Monetary
Cluster 2	Loyal Customers	573	Low Recency, High Frequency, High Monetary
Cluster 3	At Risk/Low Value	516	High Recency, Low Frequency, Low Monetary

Personalized Marketing Strategies

Segment Name	General Characteristics	Recommended Marketing Strategies
New / Occasional	Low Recency, Low Frequency, Low Monetary	Encourage repeat purchases through welcome campaigns, discounts, and personalized recommendations
Potential Loyals	High Recency, High Frequency, High Monetary	Nurture with loyalty programs, targeted offers, and upselling opportunities
Loyal Customers	Low Recency, High Frequency, High Monetary	Reward with VIP programs, exclusive deals, and retention campaigns
At Risk/Low Value	High Recency, Low Frequency, Low Monetary	Re-engagement campaigns, win-back offers, and targeted promotions

How to improve ROI?

Segment	Uplift Rate (%)	Incremental Revenue ($)	Marketing Cost ($)
At Risk/Low Value	10.0	630.31	500
At Risk/Low Value	20.0	1260.63	500
At Risk/Low Value	30.0	1890.94	500
At Risk/Low Value	50.0	3151.57	500
New / Occasional	10.0	696.50	500
New / Occasional	20.0	1393.01	500
New / Occasional	30.0	2089.51	500
New / Occasional	50.0	3482.52	500
Potential Loyals	10.0	-52.73	500
Potential Loyals	20.0	-105.47	500
Potential Loyals	30.0	-158.20	500
Potential Loyals	50.0	-263.66	500



Converting 20% of At Risk Segment to Loyal Customers behavior adds $1260 revenue and 152% of ROI.

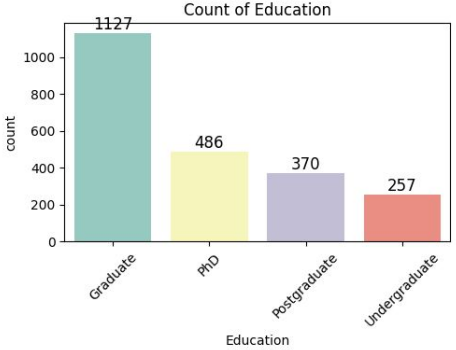
Based on assumption of $500 marketing cost for each segment.

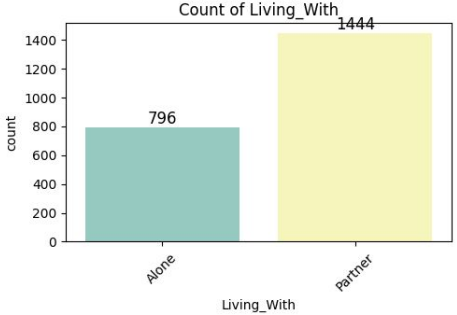
Executive Summary

- **Objective:** Analyze customer purchasing behavior to identify distinct segments
- **Methodology:** Used unsupervised machine learning (K-Means Clustering) based on RFM (Recency, Frequency, Monetary).
- **Model Optimization:** Tested multiple *K* values and evaluated them using the Silhouette Score to ensure the most distinct and accurate customer groupings.
- **Key Segments Identified:** Successfully categorized the customer base into four distinct groups: Loyal Customers, Potential Loyals, New/Occasional, and At Risk/Low Value.
- **Actionable Insights:** Developed tailored marketing strategies for each cluster - ex. VIP reward programs for loyalists and win-back campaigns for at-risk customers.
- **ROI:** Performance simulation indicates a 152% ROI if 20% of At Risk customers are converted to Loyals.

Appendix

About the data: a complete customer view: From demography to purchase behavior





Next Steps/Conclusion:

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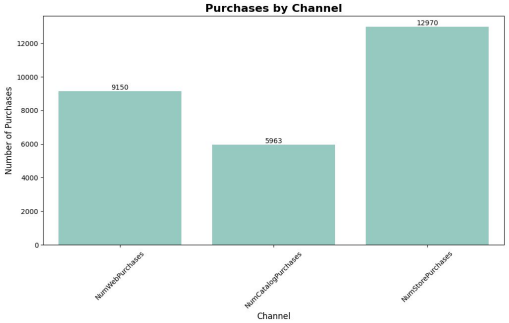
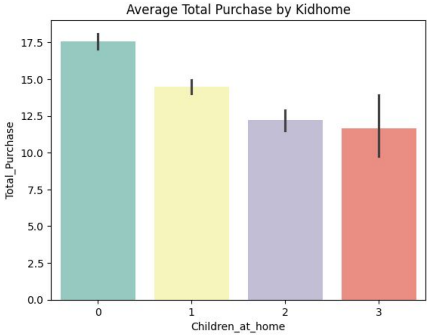
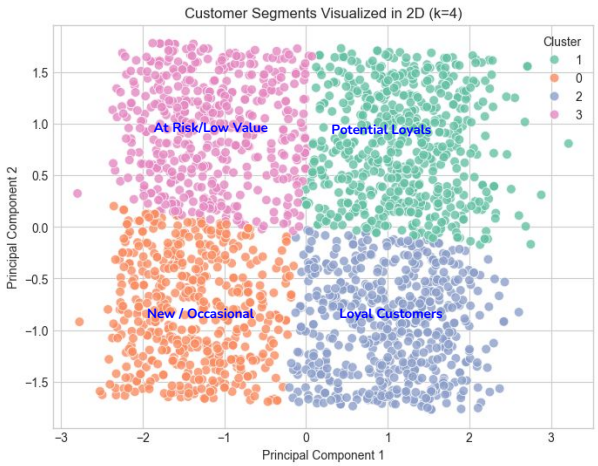




We Found Distinct Customer Clusters: RFM: Recency, Frequency, and Monetary

Method used: kMeans Clustering (unsupervised learning)



Next Steps/Conclusion:

Recommendations based on Clustering

Cluster	Segment Name	No. of Customers	General Characteristics	Marketing Insight
Cluster 0	New / Occasional	524	Low recency, Low frequency, low monetary	Encourage repeat purchases through welcome campaigns, discounts, and personalized recommendations
Cluster 1	Potential Loyals	616	High Recency, High Frequency, High Monetary	Nurture with loyalty programs, targeted offers, and upselling opportunities
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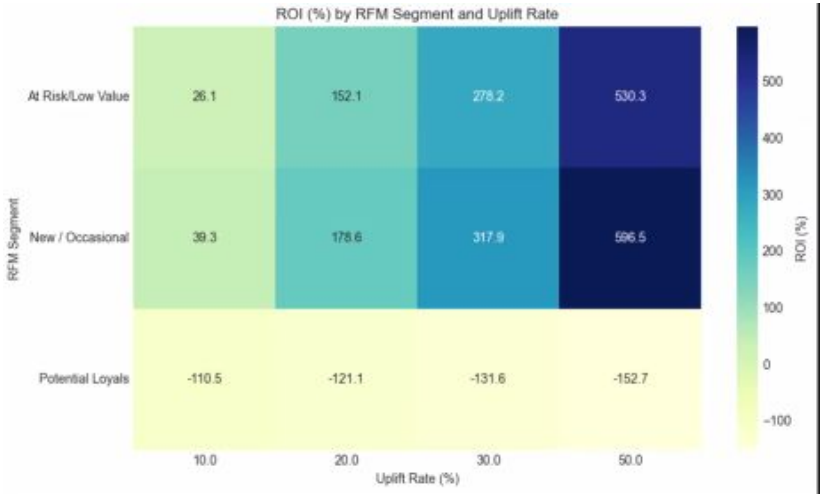
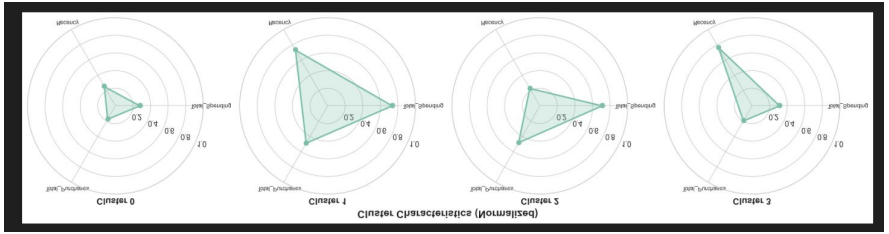
Performance Simulation - the set up:

Hypothetical marketing budget per Segment: $500

	Segment	Customers
3	Potential Loyals	616
1	Loyal Customers	573
0	At Risk/Low Value	516
2	New / Occasional	524

Heatmap of revenue uplift by segment





Performance Simulation - the results:

	Segment	Uplift Rate (%)	Incremental Revenue ($)	Marketing Cost ($)	ROI (%)
0	At Risk/Low Value	10.0	630.31	500	26.06
1	At Risk/Low Value	20.0	1260.63	500	152.13
2	At Risk/Low Value	30.0	1890.94	500	278.19
3	At Risk/Low Value	50.0	3151.57	500	530.31
4	New / Occasional	10.0	696.50	500	39.30
5	New / Occasional	20.0	1393.01	500	178.60
6	New / Occasional	30.0	2089.51	500	317.90
7	New / Occasional	50.0	3482.52	500	596.50
8	Potential Loyals	10.0	-52.73	500	-110.55
9	Potential Loyals	20.0	-105.47	500	-121.09
10	Potential Loyals	30.0	-158.20	500	-131.64
11	Potential Loyals	50.0	-263.66	500	-152.73

Based on our analysis, ABC Supermart can see 26% to almost 600% ROI if they focus on converting to At Risk/Low Value or the new/Occasional customers to Loyal customers